

Orange in Chemistry

Exercise and Implementation Companion

Complete classroom exercises, teacher implementation notes, workflows, assessment evidence, and practical delivery guidance

7 Teaching modules	72 Exercise cards	3 Delivery modes	1 Reproducible workflow culture
Orange Visual analytics	RDKit Cheminformatics engine	QSAR Models + interpretation	FAIR Provenance + reports

1. Implementation Model

Use this companion as a practical delivery manual. Each exercise can be run as a short guided lab, a homework task, or a project milestone. The recommended classroom rhythm is: predict, run, inspect, explain, report.

Phase	Teacher action	Student action	Evidence	Typical time
Prepare	Install package, test widget, open data file.	Read the prompt and predict expected chemical behaviour.	Ready workflow and notes.	5-10 min
Run	Demonstrate the core workflow once.	Execute workflow and check outputs.	Saved .ows and result table.	15-25 min
Interpret	Ask chemistry-first questions.	Explain features, structures, alerts, or model results.	Annotated table or paragraph.	15-30 min
Report	Require provenance and limitations.	Submit short reproducible report.	Methods + result + limitation.	15-45 min

Recommended delivery modes

Guided lab: Teacher demonstrates the first molecule or first model, then students complete the remaining tasks.

Inquiry lab: Students predict outcomes before running the widgets, then reconcile predictions with outputs.

Mini project: Students choose a small compound set, run a workflow, and submit a concise reproducible report.

2. Assessment Template

Criterion	Developing	Proficient	Excellent
Workflow execution	Runs with help; incomplete outputs.	Runs workflow and exports requested outputs.	Runs independently and checks edge cases.
Chemical interpretation	Names outputs but gives little chemistry.	Connects outputs to molecular features.	Explains limitations and exceptions.
Data quality	Ignores invalid or inconsistent rows.	Documents failures and preprocessing.	Proposes corrective curation strategy.
Reproducibility	Few settings recorded.	Records data, widgets, and key settings.	Includes provenance, versioning, and limitations.

Module 1. Cyclic Registry Fingerprint

Big idea	Use an interpretable 4096-bit fingerprint to connect molecular structure, cyclic motifs, and machine-learning-ready representations.
Core workflow	File -> Cyclic Registry Fingerprint -> Data Table / Matched Registry Entries
Suggested data	examples/cyclic_registry_fingerprint/ cyclic_registry_training_set.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: Drug Heterocycles

- **Learning objective:** Identify heterocycles in drug-like molecules and explain why matched registry entries matter.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Matched Registry Entries -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create a frequency table of detected heterocycle entries and annotate three molecules.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain at least two detected ring systems using chemical language.

Exercise 2: Aromatic vs Kekule

- **Learning objective:** Compare aromatic and kekulized encodings of the same ring families.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Matched Registry Entries
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Run paired SMILES examples and compare registry matches.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should recognize representation robustness and remaining edge cases.

Exercise 3: Natural Products

- **Learning objective:** Profile cyclic motifs in natural-product-like structures.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Group matched entries by family and discuss natural product diversity.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should connect motifs to compound class.

Exercise 4: QSAR/QSPR Input

- **Learning objective:** Use the 4096-bit output as QSAR features.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Test & Score

- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare a simple model using registry-enhanced features.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should separate prediction from interpretation.

Exercise 5: Feature Interpretation

- **Learning objective:** Relate high-ranking model features to matched registry entries.
- **Orange workflow:** Cyclic Registry Fingerprint -> Rank + Matched Registry Entries
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Find top-ranked bits and inspect which motifs trigger them.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should state whether a bit is interpretable or hashed.

Exercise 6: Library Comparison

- **Learning objective:** Compare motif profiles between libraries.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Matched Registry Entries
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare frequency profiles for two groups in the input table.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should present one structural difference between libraries.

Exercise 7: Docking Results

- **Learning objective:** Describe motifs among top-ranked docking candidates.
- **Orange workflow:** File -> Select Rows -> Cyclic Registry Fingerprint
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Filter top docking scores and profile cyclic motifs.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid claiming causality from docking score alone.

Exercise 8: Unwanted Motifs

- **Learning objective:** Screen for undesirable functional or structural motifs.
- **Orange workflow:** Cyclic Registry Fingerprint -> Matched Registry Entries -> Select Rows
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Filter matched entries by family/section and flag compounds.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should distinguish alerting from automatic rejection.

Exercise 9: SMARTS Learning

- **Learning objective:** Use registry hits to learn SMARTS/substructure thinking.
- **Orange workflow:** File -> Cyclic Registry Fingerprint -> Matched Registry Entries
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Predict hits before running the widget, then compare results.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain false expectations constructively.

Exercise 10: Method Development

- **Learning objective:** Use validation/collision reporting to improve an interpretable fingerprint.
- **Orange workflow:** CLI validator + Cyclic Registry Fingerprint
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Run registry validation and discuss collisions.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should propose one improvement to registry metadata.

Module 2. QSAR Widgets

Big idea	Build, validate, compare, and report QSAR models while keeping chemistry and data leakage in view.
Core workflow	File -> Mol Descriptors 2 / Fingerprint Generator -> QSAR Regression -> Test & Score
Suggested data	examples/qsar_widgets/qsar_training_set.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: Descriptor Preparation

- **Learning objective:** Generate molecular descriptors and inspect missing or constant features.
- **Orange workflow:** File -> Mol Descriptors 2 -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Identify descriptors with missing values and discuss preprocessing.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should document descriptor settings.

Exercise 2: Fingerprint QSAR

- **Learning objective:** Use binary fingerprints as model features.
- **Orange workflow:** File -> Fingerprint Generator -> QSAR Regression
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare Morgan and cyclic registry fingerprints.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain fingerprint radius/bit length tradeoffs.

Exercise 3: Simple QSAR Regression

- **Learning objective:** Train a baseline regression model for a continuous endpoint.
- **Orange workflow:** File -> Mol Descriptors 2 -> QSAR Regression -> Predictions
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Fit a model and report RMSE/MAE/R².
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should interpret metrics cautiously.

Exercise 4: Scaffold Split Validation

- **Learning objective:** Test whether performance survives scaffold-aware splitting.
- **Orange workflow:** Mol Descriptors 2 -> Scaffold Splitter -> QSAR Regression
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare random and scaffold split performance.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.

- **Assessment evidence:** Students should identify overoptimistic random-split results.

Exercise 5: Interpretable MLR

- **Learning objective:** Build an interpretable multiple-linear-regression model.
- **Orange workflow:** Mol Descriptors 2 -> MLR Model Selection
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Select few descriptors and discuss coefficients.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid overfitting and unsupported mechanistic claims.

Exercise 6: Model Comparison

- **Learning objective:** Compare descriptor, Morgan, and cyclic-registry feature sets.
- **Orange workflow:** Three feature branches -> Test & Score
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create a model comparison table.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should report both performance and interpretability.

Exercise 7: External Prediction

- **Learning objective:** Apply the model to unseen molecules.
- **Orange workflow:** Training branch + external File -> Predictions
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Predict an external set and flag questionable cases.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should connect prediction to applicability domain.

Exercise 8: Activity Cliffs

- **Learning objective:** Investigate why similar molecules can have different activity.
- **Orange workflow:** Fingerprint Generator -> Activity Cliff Finder
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Identify cliff pairs and inspect structures.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should describe SAR implications.

Exercise 9: QSAR Reporting

- **Learning objective:** Prepare a reproducible QSAR report.
- **Orange workflow:** Full QSAR workflow -> Save/Report
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Document data, preprocessing, split, model, metrics, limitations.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should include provenance and limitations.

Exercise 10: Capstone QSAR Project

- **Learning objective:** Design a small end-to-end QSAR study.
- **Orange workflow:** Standardizer -> Features -> Split -> Model -> AD -> Report

- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Complete a concise project report.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should justify every workflow choice.

Module 3. Applicability Domain

Big idea	Decide when a QSAR prediction should be trusted, questioned, or rejected.
Core workflow	Reference File + Query File -> Applicability Domain -> Data Results / AD Summary
Suggested data	examples/applicability_domain/ ad_reference_training_set.csv + ad_query_prediction_set.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: Conceptual Introduction

- **Learning objective:** Explain why a model has a domain of reliable use.
- **Orange workflow:** Training/query data -> Applicability Domain
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Classify molecules as inside/outside domain.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should distinguish model error from domain warning.

Exercise 2: Williams Leverage

- **Learning objective:** Use leverage to identify structurally influential query compounds.
- **Orange workflow:** Descriptors -> Applicability Domain
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Interpret leverage and warning thresholds.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain the Williams plot concept.

Exercise 3: kNN Distance Domain

- **Learning objective:** Use nearest-neighbour distances to assess local similarity.
- **Orange workflow:** Descriptors/Fingerprints -> Applicability Domain
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare k values and distance thresholds.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should relate distance to chemical similarity.

Exercise 4: Mahalanobis Domain

- **Learning objective:** Use multivariate distance to flag unusual compounds.
- **Orange workflow:** Descriptor table -> Applicability Domain

- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Run Mahalanobis-based domain assessment.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should discuss scaling and descriptor correlation.

Exercise 5: External Prediction Reliability

- **Learning objective:** Combine predictions with AD flags.
- **Orange workflow:** QSAR Predictions -> Applicability Domain -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Mark predictions as reliable/caution/outside domain.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should report uncertainty labels.

Exercise 6: Descriptor Choice Effect

- **Learning objective:** Show that AD depends on representation.
- **Orange workflow:** Two descriptor/fingerprint branches -> AD
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare AD flags across feature spaces.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should not treat AD as absolute truth.

Exercise 7: Fingerprint Domain Discussion

- **Learning objective:** Discuss AD in binary fingerprint space.
- **Orange workflow:** Fingerprint Generator -> Applicability Domain
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare Tanimoto-like thinking with descriptor distance.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should identify limitations.

Exercise 8: QSAR Reporting with AD

- **Learning objective:** Add AD results to model reporting.
- **Orange workflow:** Full QSAR + AD workflow
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Prepare a report paragraph for AD.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should include rule, threshold, and consequence.

Exercise 9: Outlier Triage

- **Learning objective:** Investigate why compounds are outside domain.
- **Orange workflow:** AD Results -> Data Table -> Viewer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect flagged molecules and propose actions.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should recommend remove, retrain, or caution.

Exercise 10: Capstone AD-QSAR

- **Learning objective:** Build a QSAR model with an explicit reliability layer.
- **Orange workflow:** Standardizer -> QSAR -> AD -> Report
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Deliver a prediction table with AD decisions.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should justify the decision policy.

Module 4. Molecular Standardization

Big idea	Clean molecular structures before descriptors, fingerprints, modelling, or interpretation.
Core workflow	File -> Mol Standardizer -> Data Table / downstream feature widget
Suggested data	examples/molecular_standardization/standardization_training_set.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: Why Standardization Matters

- **Learning objective:** Show that representation choices affect downstream results.
- **Orange workflow:** File -> Mol Standardizer -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare original and standardized SMILES.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should identify at least three changed records.

Exercise 2: Invalid SMILES and Sanitization

- **Learning objective:** Detect and document invalid structures.
- **Orange workflow:** File -> Mol Standardizer -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect failed rows and error/status columns.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should not silently discard failures.

Exercise 3: Salts and Mixtures

- **Learning objective:** Handle salts, mixtures, and largest-fragment choices.
- **Orange workflow:** File -> Mol Standardizer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare preserve-salts and largest-fragment logic.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain when each choice is appropriate.

Exercise 4: Charge Normalization

- **Learning objective:** Explore uncharging/reionization effects.
- **Orange workflow:** Mol Standardizer -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Track changes in charged examples.

- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should preserve chemically meaningful charges when needed.

Exercise 5: Nitro/Zwitterions/Quaternary Ammonium

- **Learning objective:** Test common valence-sensitive cases.
- **Orange workflow:** Mol Standardizer -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Check nitro and quaternary ammonium examples.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should understand why formal charges matter.

Exercise 6: Aromaticity and Kekulization

- **Learning objective:** Compare aromatic and kekulized representations.
- **Orange workflow:** Mol Standardizer -> Cyclic Registry Fingerprint
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Evaluate downstream motif matching.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should document representation effects.

Exercise 7: Tautomers and Protonation States

- **Learning objective:** Discuss what standardization does and does not solve.
- **Orange workflow:** Mol Standardizer -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare related protonation/tautomer examples.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid pretending pH was fully modelled.

Exercise 8: Before Fingerprints

- **Learning objective:** Show standardization effect on fingerprint similarity.
- **Orange workflow:** Mol Standardizer -> Fingerprint Generator
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare bit outputs before/after cleaning.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should report how preprocessing changes features.

Exercise 9: Before QSAR

- **Learning objective:** Show standardization as part of modelling workflow.
- **Orange workflow:** Mol Standardizer -> Descriptors -> QSAR
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Train a small model with clean data.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should describe preprocessing in methods.

Exercise 10: Reproducible Standardization Report

- **Learning objective:** Create a standardization audit trail.

- **Orange workflow:** Mol Standardizer -> Data Table/Report
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Write a protocol with settings and failures.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should include profile name and changed records.

Module 5. ChEMBL Bioactivity Curation

Big idea	Transform heterogeneous public bioactivity records into a transparent QSAR-ready dataset.
Core workflow	ChEMBL Browser -> Bioactivity Retriever -> Standardizer -> QSAR features
Suggested data	examples/chembl_bioactivity_curation/chembl_bioactivity_curation_demo.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: What is ChEMBL Data

- **Learning objective:** Identify entities: compound, target, assay, document, activity.
- **Orange workflow:** Demo CSV -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Label key columns and explain relationships.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should not treat all rows as equivalent measurements.

Exercise 2: Target Search and Confidence

- **Learning objective:** Use target confidence to select biologically meaningful records.
- **Orange workflow:** ChEMBL Browser -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare targets with different confidence scores.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should justify target selection.

Exercise 3: Endpoints/Units/Relations

- **Learning objective:** Curate IC50/Ki/Kd/EC50 and relation symbols.
- **Orange workflow:** Bioactivity table -> Select Rows
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Filter consistent endpoint and unit combinations.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain pChEMBL and relation qualifiers.

Exercise 4: Retrieving Bioactivities

- **Learning objective:** Download or inspect bioactivities for a target.
- **Orange workflow:** ChEMBL Browser -> Bioactivity Retriever
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create an initial raw table.

- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should preserve raw data provenance.

Exercise 5: Assay Filtering

- **Learning objective:** Filter by assay type, confidence, organism, and validity comments.
- **Orange workflow:** Bioactivity Retriever -> Select Rows
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create a curated subset.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should document inclusion/exclusion rules.

Exercise 6: Structure Cleaning

- **Learning objective:** Standardize retrieved compounds.
- **Orange workflow:** Bioactivity table -> Mol Standardizer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect failures and salts.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should connect curation and chemistry.

Exercise 7: Duplicates and Aggregation

- **Learning objective:** Handle repeated measurements for the same compound.
- **Orange workflow:** Data Table -> Group/Aggregate workflow
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare mean/median/best-case aggregation.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should state aggregation rule.

Exercise 8: QSAR-Ready Dataset

- **Learning objective:** Build a clean table for modelling.
- **Orange workflow:** Curated Bioactivity -> Descriptors/Fingerprints
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create final table with endpoint and features.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should include units and transformed endpoint.

Exercise 9: Provenance and FAIR Reporting

- **Learning objective:** Write a reproducible data curation protocol.
- **Orange workflow:** Workflow plus exported tables
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Prepare a methods-ready curation paragraph.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should report ChEMBL version/date if available.

Exercise 10: Capstone Curation Project

- **Learning objective:** Prepare a small curated dataset from a target.

- **Orange workflow:** ChEMBL -> Standardizer -> QSAR-ready table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Deliver raw, curated, and modelling-ready files.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should justify every filter.

Module 6. Scaffold and Activity Cliff Analysis

Big idea	Use scaffolds and cliffs to connect chemical series, validation design, and medicinal chemistry interpretation.
Core workflow	File -> Scaffold Analysis / Scaffold Splitter / Activity Cliff Finder
Suggested data	examples/scaffold_activity_cliff_analysis/ scaffold_activity_training_set.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: What Scaffolds Represent

- **Learning objective:** Explain scaffold as a series-level abstraction.
- **Orange workflow:** File -> Scaffold Analysis -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare molecule and scaffold columns.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid overinterpreting scaffold as full chemistry.

Exercise 2: Murcko and Generic Scaffolds

- **Learning objective:** Compare Murcko and generic scaffold definitions.
- **Orange workflow:** Scaffold Analysis -> Scaffold Summary
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect scaffold grouping differences.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should describe what atoms/bonds are abstracted.

Exercise 3: Scaffold Frequency

- **Learning objective:** Profile dominant chemical series.
- **Orange workflow:** Scaffold Analysis -> Summary
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Rank scaffolds by compound count.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should identify series imbalance.

Exercise 4: Scaffold Splitter

- **Learning objective:** Create scaffold-aware train/test splits.
- **Orange workflow:** Scaffold Splitter -> Train/Test Data
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect scaffold distribution across splits.

- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid scaffold leakage.

Exercise 5: Random vs Scaffold Split

- **Learning objective:** Compare model performance under split strategies.
- **Orange workflow:** QSAR workflow with random vs scaffold split
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Report performance differences.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain why scaffold split is often harder.

Exercise 6: Activity Cliff Concepts

- **Learning objective:** Define similar structure/different activity.
- **Orange workflow:** File -> Fingerprint Generator -> Activity Cliff Finder
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Find example cliff pairs.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should connect cliffs to SAR uncertainty.

Exercise 7: Detecting Cliffs with Fingerprints

- **Learning objective:** Tune similarity and potency thresholds.
- **Orange workflow:** Activity Cliff Finder
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Explore threshold sensitivity.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should report parameter choices.

Exercise 8: Medicinal Chemistry Interpretation

- **Learning objective:** Interpret transformations in cliff pairs.
- **Orange workflow:** Activity Cliff Finder -> Data Table -> Viewer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Discuss substituent changes and activity shifts.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should avoid unsupported mechanism claims.

Exercise 9: Scaffold-Cliff-QSAR-AD Link

- **Learning objective:** Connect cliffs to validation and domain warnings.
- **Orange workflow:** Scaffold + Cliff + QSAR + AD workflow
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Identify difficult regions of chemical space.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should synthesize multiple outputs.

Exercise 10: Capstone Scaffold/Cliff Project

- **Learning objective:** Analyze a chemical series and report cliffs.

- **Orange workflow:** Full scaffold/cliff workflow
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Deliver scaffold summary and selected cliff examples.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should propose next design decisions.

Module 7. Supplementary Cheminformatics Workflows

Big idea	Use core cheminformatics widgets for similarity, substructure search, diversity, library design, visualization, MMPA, reactions, and QC.
Core workflow	Similarity Search / Substructure Search / Diversity Picker / MMP / Reaction Enumerator / Viewers
Suggested data	examples/supplementary_cheminformatics_workflows/*.csv

Implementation sequence

Before class: install/update package, open the example CSV, run the first workflow, and confirm expected outputs.

During class: start with a prediction question, run the widget, inspect both numerical and chemical outputs, then require a short explanation. **After class:** collect the .ows workflow, exported table, and one-paragraph interpretation with limitations.

Exercise cards

Exercise 1: Similarity Search

- **Learning objective:** Find molecules similar to a query and interpret Tanimoto scores.
- **Orange workflow:** File -> Fingerprint Generator -> Similarity Search
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Rank library molecules by similarity.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should state fingerprint and threshold.

Exercise 2: Substructure SMARTS Search

- **Learning objective:** Search a library for a structural motif.
- **Orange workflow:** File -> Substructure Search -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Use SMARTS examples and inspect matches.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain SMARTS specificity.

Exercise 3: Diversity Picker

- **Learning objective:** Select a small diverse subset from a larger library.
- **Orange workflow:** File -> Fingerprint Generator -> Diversity Picker
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Pick a representative set and justify size.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should distinguish diversity from quality.

Exercise 4: Drug Filter/Likeness

- **Learning objective:** Apply drug-like or rule-based filters.
- **Orange workflow:** File -> Drug Filter -> Data Table

- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare accepted/rejected compounds.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should treat filters as heuristics.

Exercise 5: SDF I/O and Provenance

- **Learning objective:** Read/write molecule files while keeping metadata.
- **Orange workflow:** SDF Reader -> Processing -> SDF Writer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Export processed structures with properties.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should verify metadata survives.

Exercise 6: Molecule Visualization

- **Learning objective:** Use viewers for structural quality control.
- **Orange workflow:** File -> Mol Viewer / Mol 3D Viewer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect problematic entries visually.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should identify visual QC limitations.

Exercise 7: R-Group Decomposition

- **Learning objective:** Analyze substituent patterns in a series.
- **Orange workflow:** File -> R-Group Decomposition -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Map substituents to a common core.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should interpret R-groups chemically.

Exercise 8: Matched Molecular Pairs

- **Learning objective:** Find transformations associated with property changes.
- **Orange workflow:** File -> Matched Molecular Pairs
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Review transformations and activity deltas.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should connect MMPs to design hypotheses.

Exercise 9: Reaction Enumeration

- **Learning objective:** Generate virtual products from rules and reactants.
- **Orange workflow:** Reactants + Rules -> Reaction Enumerator
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Inspect enumerated products and failures.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should document reaction scope assumptions.

Exercise 10: Pharmacophore Fingerprint Search

- **Learning objective:** Search by pharmacophore-like feature patterns.
- **Orange workflow:** File -> PharmaFP Search -> Data Table
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Compare structural vs pharmacophoric similarity.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should explain feature abstraction.

Exercise 11: Integrated QC Pipeline

- **Learning objective:** Design a reliable preparation pipeline for a library.
- **Orange workflow:** Standardizer -> Filters -> Fingerprints -> Viewer -> Writer
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Create a QC table and export accepted compounds.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should document every exclusion.

Exercise 12: Capstone Library Design

- **Learning objective:** Design a focused screening library.
- **Orange workflow:** Multiple workflow branches
- **Implementation steps:** Load the suggested data; connect widgets in the listed order; run the workflow; inspect the main output and at least one diagnostic/interpretive output; save the workflow and export the result table. Specific task: Deliver selected set and design rationale.
- **Student deliverable:** Saved .ows workflow, exported result table, and 5-8 sentence interpretation.
- **Assessment evidence:** Students should balance diversity, motifs, and constraints.

Quick Implementation Checklists

Installation check

- `conda activate owcheminf`
- `pip install -e .`
- `orange-canvas`
- Confirm widget category: Cheminf - Processing

Before each lesson

- Open the dataset from examples/...
- Run the workflow once as teacher
- Prepare expected output screenshots or result snippets
- Decide assessment evidence: table, report, or workflow

Student submission package

- Orange .ows workflow
- Exported CSV or table screenshot
- Short methods paragraph with settings
- Short interpretation with limitations

Troubleshooting

- If widget is missing: reinstall editable package
- If SMILES fail: inspect standardization/failure columns
- If model looks too good: check duplicates and split leakage
- If outputs differ: record package/RDKit versions